Meeting Summaries of June 11

Date: June 11, 2026
Time: PDT 7:00 to 7:30pm
Attendees: Prof. Jan Ciborowski, Dr. Yue Zhang and Feng Gu
Topic: Reproducing earlier analyses, checking contamination-score methods, Chapter 2 writing, and follow-up on quantile regression.

Questions, Feedbacks and Suggestions

- Prof. Jan: Feng should first **reproduce the original analysis**, including PCA and Bray-Curtis polar ordination, before arguing that the newer workflow is an improvement.
- Prof. Jan: The current **NMDS ordination** does not directly match the original polar-ordination target that covers all sites within the 0 to 1 poles-interval, it prefers to update the current methods to make a clipped ordination segment that covers all rest sites as in the original study.
- Prof. Jan: The **PCA contamination scores** should be checked using standardized PC scores, consistent with the earlier thesis, rather than relying only on the current raw-score combination. But it is worth to check 'if the PC scores were alread mapped from log-transformed standardized chemical data, what does it mean to re-standardize the PC scores again?'

Methodological priority

Before extending the method, Feng needs to show that he can **duplicate the earlier workflow and results**. Otherwise it is difficult to judge whether the new results reflect real improvement or unresolved differences in data processing. Feng returned to the earlier 'unit-mismatch' issue in the two daughter chemical datasets. One comes with 255 samples and most are after 1991 survey, and the other comes with all samples in 1991 survey. Feng keeps a bit skeptical about was the 'unit-mismatch' issue fixed in 2008 work, as he could not deplicate the same PCA loading table as in the earlier thesis. Furthermore, Feng did not read it from previous work what specific units were used before as the chemicals were stored in various units in the given datasets. It is important to shape the PC loading results, as there were log-transformation and standardization steps before PCA. Various units will determine the magnitude of log-transformation in falttening the recorded values, and finally decide the covariation structure that PCA will be applied on.
(Note: standardize variable and then apply PCA on scaled covariance matrix is equivalent to PCA on correlation matrix, as the correlation coefficient is the covariance of standardized variables.)

- Prof. Jan: The fluctuation of confidence intervals in **piecewise quantile regression** is not only a problem but also part of what the method reveals. Feng should document how this behaves across different quantiles.
- Prof. Jan: Feng should keep writing **Chapter 2** while the methodological checks continue, upload the latest PDF, and compare the writing style with previous theses from the lab and TRU.

Action Items

- Feng will upload the latest **Chapter 2 PDF** to the website and email Prof. Jan.
- Feng will standardize the **PC scores** and compare the resulting contamination scores with the current approach.
- Feng will create **3D ordination plots** to inspect the third axis in the NMDS result.
- Feng will organize the **piecewise quantile-regression** results for the 20th, 50th, and 80th percentiles, including the behavior of confidence intervals.
- Prof. Jan will review the uploaded Chapter 2 draft and continue checking the remaining **chemical-data discrepancies**.

Summary of Discussion

- Feng reported recent progress in **writing Chapter 2** and in revising the environmental-analysis workflow. The meeting started with a discussion of delayed feedback, the current writing status, and the need to upload a newer PDF for review.
- Feng explained the current use of **NMDS with Bray-Curtis similarity** for biota composition analysis. Prof. Jan responded that the present method should be compared

against the original workflow based on Bray-Curtis polar ordination with defined end poles, because that was the basis of the earlier interpretation.

- The discussion then moved to **method reproducibility**. Prof. Jan emphasized that without reproducing the earlier PCA tables and ordination logic, it is difficult to assess whether the current changes represent improvement or simply a different analytical path.

- Feng also described possible use of intermediate results from the earlier **CSS-related analysis** to reproduce clustering and classification. During this discussion, both of them noted that the remaining chemical-data discrepancies might involve more than simple unit mismatches.

- The meeting included discussion of **piecewise quantile regression**. Feng reported that the 50th-percentile result was available, while the 20th and 80th percentiles still showed unstable confidence intervals. Prof. Jan explained that this instability is tied to the assumptions of quantile-based analysis and should be documented rather than hidden.

- They also discussed **PCA score construction**, including raw versus standardized PC scores and the weighting of components. Prof. Jan advised Feng to compare the standardized approach used in the earlier thesis, and Feng agreed to prepare that comparison.